Downstream Routing Benchmark: Hazard Avoidance vs Travel-Time Overhead (λ = 5.0)

Evaluated Across 150 Representative Origin-Destination Trips in Hai Chau District

Routing Strategy Results

(Flooded length, Time overhead)

- **Shortest time**
  1063.6 m | +0.0% time
- **LR**
  780.7 m | +4.1% time
- **SVM (RBF)**
  692.6 m | +11.6% time
- **Random Forest**
  683.6 m | +7.6% time
- **XGBoost**
  639.8 m | +13.5% time
- **MLP**
  625.8 m | +16.5% time
- **GraphSAGE**
  689.5 m | +18.9% time
- **Oracle (simulated)**
  485.6 m | +21.9% time